

Elevon Direct-Drive Pin Calculation

Purpose: preliminary calculation of the aerodynamic elevon load, hinge moment, direct-drive torque, V-tail force components, and minimum solid pin size. This file does not select a servo model; it only defines the mechanical load basis for later actuator selection.

Flight-control context: the uploaded Fundamentals of Flight Control slides identify control surfaces such as ailerons, elevators, rudders, and elevons as the surfaces that generate aircraft control moments. For actuator sizing, the relevant local quantity is the control-surface hinge moment. The same actuator-sizing logic used in the aileron calculation is applied here to one V-tail elevon.

1. Input values and assumptions

Property	Symbol	Value	Unit	Comment
Air density	ρ	1.17	kg/m ³	From aircraft sizing table
Sizing speed	V	24.0	m/s	Cruise speed used as current sizing speed
Stall speed	V _{stall}	14.0	m/s	Only used for comparison
Kinematic viscosity	ν	1.44e-05	m ² /s	From aircraft sizing table
Wing mean chord	c _w	0.300	m	Used only for Reynolds-number check
V-tail angle to horizon	θ	37.9	deg	From screenshot
Tail moment arm	L _{tail}	0.90	m	HT/VT moment arm from screenshot
Total V-tail area	S _{V-tail}	0.2452	m ²	From screenshot
Full V-tail span	b _{V-tail}	0.990	m	From screenshot
V-tail chord	c _{V-tail}	0.248	m	From screenshot
Single elevon chord	c _e	0.099	m	From screenshot
Single elevon span	b _e	0.495	m	Assumed one elevon covers one half of V-tail span
Center-of-pressure arm	x _{cp}	0.0396	m	Assumed 0.40*c _e behind hinge
Safety factor	SF	3.0	-	Preliminary mechanical design factor
Drive efficiency	η	0.90	-	Direct pin/coupling efficiency estimate
Pin length	L _{pin}	30.0	mm	User-proposed total pin length

Important assumption: because the screenshot leaves the single elevon span blank but notes that the elevon spans almost the whole V-tail surface, one elevon is assumed to cover one V-tail half-panel. Therefore $b_e = b_{V\text{-tail}}/2 = 0.495$ m. If the final elevon span is shorter, the aerodynamic force and hinge moment scale approximately linearly with b_e .

2. Basic aerodynamic quantities

$$q = 0.5 * \rho * V^2 = 0.5 * 1.17 * 24.0^2 = 336.96 \text{ Pa}$$

$$q_{\text{stall}} = 0.5 * \rho * V_{\text{stall}}^2 = 114.66 \text{ Pa}$$

$$q/q_{\text{stall}} = 2.94$$

Therefore the aerodynamic loads at 24 m/s are about 2.94 times the loads at 14 m/s. If the aircraft can fly faster than 24 m/s, all aerodynamic forces and moments must be multiplied by $(V_{\text{max}}/24)^2$.

$$Re = V * c_w / \nu = 24.0 * 0.30 / 1.44e-05 = 500000$$

$$A_e = b_e * c_e = 0.495 * 0.099 = 0.04900 \text{ m}^2$$

$$x_{cp} = 0.40 * c_e = 0.40 * 0.099 = 0.0396 \text{ m}$$

3. Elevon aerodynamic normal force

The preliminary elevon normal-force coefficient is estimated with $C_{N,e} = 2\pi\tau\delta_e$, where δ_e is in radians. The lower estimate uses $\tau = 0.5$. The conservative estimate uses $\tau = 1.0$.

Elevon deflection	$C_{N,e}$ lower	$C_{N,e}$ conservative	$F_{N,e}$ lower [N]	$F_{N,e}$ conservative [N]
10 deg	0.548	1.097	9.05	18.11
15 deg	0.822	1.645	13.58	27.16
20 deg	1.097	2.193	18.11	36.22

4. Hinge moment and direct-drive torque

The formal hinge-moment expression is $HM_e = c_h * q * A_e * c_e$. Since the exact hinge-moment coefficient c_h is not available for the present V-tail/elevon geometry, the force-arm approximation is used for preliminary sizing: $HM_e = F_{N,e} * x_{cp}$.

$$HM_e = F_{N,e} * x_{cp}, \text{ with } x_{cp} = 0.0396 \text{ m}$$

For a coaxial direct-drive pin, there is no horn/pushrod ratio. The pin torque is approximately the elevon hinge moment. The design torque is $T_{\text{design}} = SF * HM_e / \eta$.

$$T_{\text{design}} = SF * HM_e / \eta = 3.0 * HM_e / 0.90$$

Elevon deflection	HM_e lower [Nm]	HM_e conservative [Nm]	T_{design} [Nm]
10 deg	0.359	0.717	2.390
15 deg	0.538	1.076	3.585
20 deg	0.717	1.434	4.781

Worst conservative case in this calculation: $\delta_e = 20$ deg, $HM_e = 1.434$ Nm, and $T_{\text{design}} = 4.781$ Nm.

5. V-tail force components and aircraft-level moments

The following projection is approximate and is used only to understand aircraft-level pitch/yaw authority. It is not the servo-sizing quantity. For a V-tail surface angle θ measured from the horizon, the normal force on one panel is split into vertical and lateral components as $F_{\text{vertical}} = F_{N,e}\cos(\theta)$ and $F_{\text{side}} = F_{N,e}\sin(\theta)$.

$$\cos(\theta) = \cos(37.9 \text{ deg}) = 0.789; \sin(\theta) = 0.614$$

For symmetric elevon deflection, the lateral side-force components largely cancel and the vertical components add, producing an approximate pitch moment. For differential/ruddervator deflection, the vertical components largely cancel and the side-force components add, producing an approximate yaw moment.

Deflection	Vertical component one side [N]	Side component one side [N]	Symmetric pitch moment [Nm]	Differential yaw moment [Nm]
10 deg	14.29	11.12	25.72	20.02
15 deg	21.43	16.69	38.58	30.03
20 deg	28.58	22.25	51.44	40.05

These pitch/yaw moments are useful for control-authority checks, but the servo and direct-drive pin must be sized from the hinge moment, not from the aircraft-level pitch or yaw moment.

6. Solid pin diameter in torsion

The direct-drive pin is assumed to be a solid circular metal pin transmitting the worst conservative design torque. The torsional shear stress is calculated with $\tau_{\max} = 16T/(\pi d^3)$.

$$T = T_{\text{design,max}} = 4.781 \text{ Nm}$$

$$\tau_{\max} = 16T/(\pi d^3)$$

$$d_{\min} = (16T/(\pi \tau_{\text{allow}}))^{1/3}$$

Allowable shear stress [MPa]	Required solid pin diameter [mm]
50	7.87
100	6.24
150	5.45
200	4.96

For an allowable shear stress of 100 MPa, the mathematical minimum diameter is 6.24 mm. Since 6 mm would exceed 100 MPa in this conservative case, the minimum practical standard diameter is 7 mm. An 8 mm pin gives a much better stiffness and stress margin.

7. Pin stress and twist for common diameters

Pin twist is calculated with $\theta_{\text{twist}} = T \cdot L / (J \cdot G)$, where $J = \pi \cdot d^4 / 32$. Twist matters for control precision even when the pin does not fail structurally.

Pin diameter [mm]	Torsional shear stress [MPa]	Twist aluminium [deg]	Twist steel [deg]
3	901.8	39.74	13.08
4	380.4	12.58	4.14
5	194.8	5.15	1.70
6	112.7	2.48	0.82
7	71.0	1.34	0.44
8	47.6	0.79	0.26

A 6 mm pin is close to the lower boundary for this elevon calculation. A 7 mm pin brings the shear stress below 100 MPa and reduces twist. An 8 mm pin is mechanically preferable if there is sufficient space.

8. Bearing pressure check

If the direct-drive pin also carries radial load at the bearing, the bearing pressure can be estimated with $p = F / (d \cdot L_b)$. This is a simplified bearing-pressure check, not a detailed bearing-life calculation.

$$F_{\text{design}} = SF \cdot F_{N,e,\text{max}} = 3.0 \cdot 36.22 = 108.65 \text{ N}$$

$$\text{For } d = 7 \text{ mm and } L_b = 6 \text{ mm: } p = 108.65 / (0.007 \cdot 0.006) = 2.59 \text{ MPa}$$

This pressure is not high for a proper metal bearing or bushing. In printed plastic supports, the more critical risks are ovalization, wear, misalignment, and local cracking.

9. Minimum pin length

The torsional stress of a circular pin does not increase with length. Length mainly affects twist and the amount of mechanical engagement. Therefore the minimum length is governed by connection design, not by torsional strength alone.

$$L_{\text{pin}} \geq L_{\text{servo_coupling}} + L_{\text{wing_bearing}} + \text{gap} + L_{\text{aileron_torque_insert}}$$

For a 7 mm pin, the aileron-side torque engagement should be at least 2d to 3d, i.e. approximately 14 to 21 mm, unless a keyed, splined, clamped, or cross-pinned connection is used and verified separately. The wing bearing contact length should be at least approximately d, i.e. about 7 mm.

Your proposed 30 mm total pin length is therefore a reasonable minimum practical length, but it leaves limited room for coupling, bearing support, gap clearance, and a strong aileron torque insert. It should not be reduced unless the detailed CAD connection proves sufficient engagement and alignment.

10. Design conclusion for the current elevon

Result	Value
Single elevon area	0.04900 m ²
Worst conservative normal force at 20 deg and 24 m/s	36.22 N
Worst conservative hinge moment at 20 deg and 24 m/s	1.434 Nm
Direct-drive design torque with SF=3 and eta=0.90	4.781 Nm
Theoretical pin diameter at tau_allow=100 MPa	6.24 mm
Practical minimum standard pin size	7 mm diameter x 30 mm length
Mechanically preferred pin size if space allows	8 mm diameter x 30 mm length

Final recommendation for the calculation file: use 7 mm x 30 mm as the practical minimum metal pin for the current elevon calculation. If the geometry allows it, use 8 mm x 30 mm because it reduces torsional shear stress and twist substantially. The pin must be fixed to the elevon through a torque insert, clamp, key, spline, cross-pin, or equivalent feature; it must not simply rotate freely inside the elevon.

If the final maximum airspeed is higher than 24 m/s, multiply $F_{N,e}$, HM_e , T_{design} , pitch moment, and yaw moment by $(V_{max}/24)^2$ and repeat the pin sizing.